

ITEP 414 – SYSTEM ADMINISTRATION AND MAINTENANCE

ENTERPRISE INFRASTRUCTURE PLAN

ABC Startup Solutions

Week 2

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Program	Bachelor of Science in Information Technology
Course	ITEP 414 – System Administration and Maintenance
Project Type	Individual Portfolio Project
Company Size	20 Employees
Office	Santa Cruz, Laguna, Philippines

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Executive Summary

This Enterprise Infrastructure Plan defines the first technology environment for ABC Startup Solutions, a hypothetical software solutions company that employs 20 people and operates in a single floor office space. As it stands right now, the company doesn't have any computer systems, servers, network infrastructure, Internet infrastructure, and security policies in place.

The suggested architecture would include centralizing network connectivity, firewall-secured internal network, switch management, wireless network connections, dedicated

server hardware, NAS storage, UPS protection, and backup devices. The suggestions have been made in consideration of a small-scale startup but will provide room for expansion as well.

Part 1

Company Profile

Company Name: ABC Startup Solutions

Nature of Business: Software development, web and mobile application development, IT consulting, and managed software services.

Office Location: Santa Cruz, Laguna, Philippines (fictional office location for academic planning purposes)

Vision: To become a trusted technology partner that delivers secure, innovative, and user-centered digital solutions for growing organizations.

Mission: To provide reliable software and technology services through skilled people, effective processes, secure infrastructure, and continuous innovation.

Organizational Structure

Department	Employees	Primary Responsibilities
Information Technology	5	Infrastructure, development, technical support, security, and system administration
Human Resources	4	Recruitment, employee records, benefits, and personnel administration
Finances	5	Accounting, budgeting, payroll, financial records, and reporting
Sales	6	Client acquisition, sales operations, account management, and customer relations
Total	20	Company-wide operations

Initial Infrastructure Assumptions

The company occupies one office floor and needs a centralized, secure, and manageable network. Most employees use fixed workstations, while selected IT personnel and managers require laptops for mobility. Shared services such as file storage, printing, backups, and internal applications should be accessible through the office network

Part 2

Enterprise Hardware Inventory

The hardware plan is sized for 20 employees while retaining a small amount of capacity for growth, shared services, and infrastructure management.

Asset ID	Hardware	Qty.	Department/Location	Purpose/Justification
HW-001	Desktop Computer	16	HR, Finance, Sales, IT	Primary fixed workstations for employees who perform daily office and development tasks
HW-002	Business Laptop	4	IT / Management	Mobility for IT administrators and personnel who require flexible work locations.
HW-003	Server	1	IT / Server Room	Hosts internal services, file sharing, application services, and centralized administration.
HW-004	Managed Gigabit Switch	1	Network Rack	Central wired connectivity for workstations, server, printer, NAS, and APs.
HW-005	Enterprise Router	1	Network Rack	Routes traffic between the office network and the ISP connection.
HW-006	Network Printer	2	Shared Office Areas	Shared printing for departments without requiring a printer at every desk.
HW-007	UPS	3	Server/ Network Rack / NAS	Protects critical equipment from short outages, surges, and unsafe shutdowns.
HW-008	Wireless Access Point	2	Office Ceiling	Provides reliable wireless coverage and supports mobile devices.
HW-009	NAS Storage	1	Server Room	Centralized file storage and backup repository
HW-010	External Backup Drive	2	IT / Secure Storage	Offline backup rotation to reduce dependence on continuously connected storage.
HW-011	Monitor	20	Employee Workstations	Dedicated display for each workstation

Selection rationale: Sixteen desktops cover standard fixed workstations, while four laptops provide mobility for IT and management. A 24-port managed switch is

recommended even though only part of its capacity is initially used, leaving room for additional endpoints, printers, cameras, or other networked devices.

Part 3

Enterprise Software Inventory

Software is selected based on workstation productivity, software development, system administration, virtualization, security, and remote support requirements.

Software	Version	License	Purpose
Windows 11 Pro	Current supported release	Commercial	Primary operating system for employee workstations; supports business management and security features.
Ubuntu Server	24.04 LTS	Open Source	Stable server operating system for internal services, development, and server administration.
Microsoft Office/ Microsoft 365	Current supported release	Commercial / Subscription	Documents, spreadsheets, presentations, email, and business productivity.
Visual Studio Code	Current supported release	Free / Open Source	Lightweight development environment for programming and configuration work
Git	Current supported release	Open Source	Source-code version control and collaboration
GitHub Desktop	Current supported release	Free	Graphical Git workflow for repository management.
VirtualBox	Current supported release	Open Source	Virtual machine testing and practice environments.
Google Chrome	Current supported release	Free	Web browsing, web application testing, and cloud service access
Microsoft Defender	Built-in	Included	Endpoint antivirus and threat protection on Windows systems.
AnyDesk	Current supported release	Commercial / Free options	Authorized remote support and troubleshooting

7-zip	Current supported release	Open Source	Compression, extraction, and archive management.
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Software administration should follow licensing requirements, approved software lists, patch management procedures, and least-privilege principles. Unsupported or unnecessary applications should not be installed on company devices.

Part 4

Enterprise Network Inventory

The network is designed around a centralized, structured cabling approach. Critical equipment is placed in a secured network rack/server area, while user devices connect through the managed switch or authorized wireless access points.

Network Equipment	Qty.	Recommended Specification	Purpose
ISP Modem / ONT	1	Business-class ISP device	Terminates the Internet service provided by the ISP
Enterprise Router	1	Gigabit-capable, VPN/QoS capable	Provides routing, DHCP/NAT functions as appropriate, and controlled WAN connectivity
Firewall	1	Next-generation firewall	Filters traffic, provides security policies, and creates a protected boundary for the internal network
Managed Switch	1	24-port Gigabit, VLAN capable	Central wired connectivity and logical network segmentation.
Wireless Access Point	2	Business Wi-Fi, VLAN capable	Wireless coverage for employees and future guest access.
Patch Panel	1	24-port CAT6	Terminates and organizes structured Ethernet cabling.
CAT6 Ethernet Cable	305 m	CAT6 solid copper	Structured horizontal cabling for workstations and network devices
RJ45 Connectors	50	CAT6 compatible	Cable termination and maintenance stock.

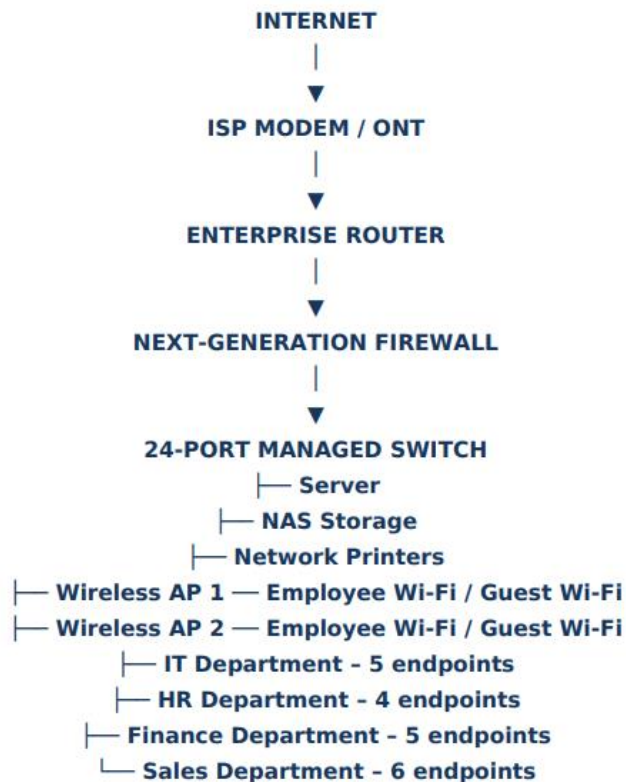
Rack Cabinet	1	Wall/floor mounted network rack	Secures and organizes active and passive network equipment.
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Suggested logical segmentation: VLAN 10 – IT, VLAN 20 – HR, VLAN 30 – Finance, VLAN 40 – Sales, VLAN 50 – Servers, and VLAN 60 – Guest Wi-Fi. Segmentation reduces unnecessary lateral access and provides a foundation for security policies.

Part 5

Enterprise Network Diagram

The following logical topology represents the proposed enterprise network. The security boundary is placed between the router and internal LAN so that traffic entering or leaving the office can be controlled by firewall policies.



Draw.io deliverable: The final GitHub submission should include the editable .drawio file plus exported PNG and PDF versions. Use standard network symbols, directional/logical connectors, labels, and a title containing the student's name.

Part 6

System Administration Roles

Helpdesk Technician

Responsibilities: Provides first-line technical support for users, including troubleshooting hardware, operating systems, applications, accounts, printers, and basic connectivity. The helpdesk records incidents, prioritizes requests, follows documented procedures, and escalates complex problems.

Skills: Customer service, troubleshooting, Windows administration, basic networking, ticket management, communication, documentation.

Common Tools: Ticketing systems, remote-support tools, Windows tools, Microsoft 365 administration portals, knowledge bases.

Certifications: CompTIA A+, ITIL Foundation, Microsoft fundamentals certifications.

Network Administrator

Responsibilities: Designs, configures, monitors, and maintains routers, switches, firewalls, wireless networks, VLANs, VPNs, and network services. The role also investigates connectivity and performance problems.

Skills: TCP/IP, subnetting, VLANs, routing, switching, wireless, firewall administration, network monitoring. **Common Tools:** Cisco IOS, firewall consoles, Wireshark, network monitoring platforms, SSH, SNMP. **Certifications:** Cisco CCNA, CompTIA Network+, vendor-specific firewall certifications.

Linux System Administrator

Responsibilities: Maintains Linux servers, user accounts, permissions, services, packages, storage, logs, backups, and system security. This role ensures server services remain available and properly maintained.

Skills: Linux command line, shell scripting, permissions, services, storage, networking, troubleshooting, security.

Common Tools: Ubuntu Server, SSH, systemd, Bash, package managers, logs, virtualization platforms. Certifications: Linux Foundation certifications, Red Hat certifications, CompTIA Linux+.

Cloud Administrator

Responsibilities: Manages cloud resources such as virtual machines, storage, networking, identity, monitoring, backups, and security controls. The role supports migration and scalable infrastructure.

Skills: Cloud architecture, IAM, virtual networks, storage, monitoring, security, automation, cost management.

Common Tools: AWS Console/CLI, Microsoft Azure Portal/CLI, cloud monitoring tools, Terraform or other infrastructure-as-code tools.

Certifications: AWS Certified Cloud Practitioner, AWS Certified Solutions Architect – Associate, Microsoft Azure Fundamentals.

How the Four Professionals Work Together

These roles form a coordinated support and infrastructure team. The Helpdesk Technician is normally the first point of contact and resolves common user issues or escalates them. The Network Administrator maintains connectivity and network security, while the Linux System Administrator manages server-side operating systems and services. The Cloud Administrator manages cloud-hosted resources and supports scalable services. They collaborate through ticketing, change-management procedures, monitoring, ITEP 414 – System Administration and Maintenance | Week 2 Page 9 documentation, and escalation so that incidents are handled efficiently without compromising security or availability.

Part 7

Infrastructure Recommendations

Internet Provider

Use a business-grade fiber Internet service with a service-level agreement, static IP option if required, adequate upload speed, and responsive technical support. A target of at least 200–500 Mbps symmetrical service is reasonable for a 20-person software company, with the exact plan selected based on actual usage and budget.

Server Specifications

Use a business-class server with at least an 8-core modern CPU, 32 GB ECC RAM, redundant storage configured with RAID 1 or RAID 10 depending on workload, dual network interfaces, and remote management. Ubuntu Server LTS can host internal services, while virtualization can separate workloads where appropriate

Backup Strategy

Follow the 3-2-1 principle: maintain at least three copies of important data, on two different media, with at least one copy stored offline or off-site. Use the NAS for centralized backups, rotate external backup drives, and periodically test restoration rather than assuming backups are usable.

Security Recommendations

Deploy a next-generation firewall, separate departments with VLANs, restrict management interfaces, disable unused ports/services, use least privilege, enable MFA for important accounts, patch systems regularly, encrypt sensitive data where appropriate, and maintain an incident-response procedure.

Antivirus / Endpoint Protection

Enable Microsoft Defender on Windows endpoints and keep security intelligence and operating-system updates current. For a growing company, evaluate centralized endpoint detection and response (EDR) when budget and risk justify it.

Password Policy

Require unique passwords, prohibit password sharing, use MFA for privileged and cloud accounts, maintain separate administrative accounts, and use a password manager approved by the organization. Avoid arbitrary frequent password changes unless there is evidence of compromise or organizational policy requires them.

Expansion Plan

Reserve switch ports, rack space, IP address capacity, and wireless capacity for growth. The 24-port switch should be replaced or expanded when utilization approaches operational limits. Future phases may add redundant Internet connectivity, additional servers, centralized identity management, cloud services, endpoint management, and stronger monitoring.

Business Continuity Considerations

Critical networking and server equipment should be protected by UPS devices and connected to appropriately rated electrical circuits. The company should document

recovery priorities, backup ownership, emergency contacts, and restoration procedures. Periodic recovery exercises are recommended so that staff can validate whether the documented procedures work under realistic conditions.

Personal Reflection

Through this project, I was able to see that system administration starts way before the computer is set up or the network cable is attached. I have come to realize that the administrator has to first have knowledge about the organization, its divisions, the number of people, their needs, and the possible threats that may impact the business. The process of making the company profile and division of the employees made me see how the business affects the technical decisions. A small business needs proper planning, as even a twenty person business becomes hard to manage once other things are introduced.

One of the most difficult aspects of this assignment was creating the enterprise network topology and finding out how the different pieces of the infrastructure would interact with one another. Simply putting a router and a switch into my drawing was not enough; I had to think about what functions the firewall, the server, the NAS, the wireless access points, printers, and other departments' endpoints would perform. The concept of VLANs was also helpful in understanding how networks could be logically designed in order to enhance their security and administration.

I found out that planning is necessary prior to deployment since planning will influence the decisions on infrastructure which would have an impact on the cost, security, performance, reliability, and future scalability of the system. In case the equipment is bought without taking into account the capacity, compatibility, back-ups, and security considerations, it could lead to costly modifications in the future. The planning will also serve as a guide for the administrator to use in troubleshooting and to make it easier for the management to know why resources are needed.

The assignment has helped me to develop skills in requirement analysis, documentation of infrastructure, choosing of appropriate technology, and professional communication as part of System Administration. This activity also made me look at the organization as an IT environment rather than separate PCs. As for further projects, I am going to improve my skills in Linux administration, networking, cloud computing, cyber security, and automation. In conclusion, I have learned from the assignment that System Administration is not only about resolving issues but also preventing them.

End of Enterprise Infrastructure Plan